GRE Quantitative Reasoning

모의고사 Set 8 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 8 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

1326. If the sum of x and 5 is divided by y, the result is 8.
Quantity A: x
Quantity B: y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

481. k and n are both positive even integers
Quantity A: The remainder when $k^2 + n$ is divided by 2
Quantity B: The remainder when k^n is divided by 2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

1248. Quantity A: The number of positive integers that are factors of both 24 and 30 but are not factors of 20
Quantity B: The number of positive integers that are factors of both 20 and 30 but are not factors of 24

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

920. Of the 7 balls in an urn, exactly one is red. Balls are to be selected from the urn one at a time, randomly and without replacement, until the red ball is selected. After the red ball is selected, no more balls will be selected.
Quantity A: The probability that a total of 3 balls will be selected
Quantity B: The probability that a total of 4 balls will be selected

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

1325. If x, y, and z are integers greater than 1 and xyz=231, what is the value of x+y+z?
A.15
B.19
C.21
D.23
E.It cannot be determined from the information given.

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

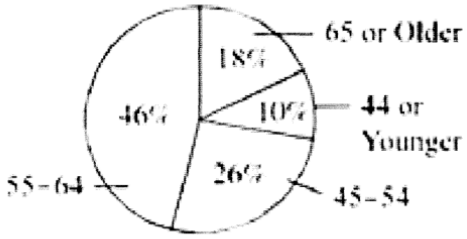
Questions 6 to 8 are based on the following data.

Survey Results on Long-Term-Care Insurance

Number of People Surveyed, by Age-Group

Age-Group (years)	Number Surveyed
44 or younger	1,600
45–54	3,500
55–64	3,100
65 or older	1,450
Total	9,650

Distribution of People Surveyed Who Had Long-Term-Care Insurance, by Age-Group (years)



Total: 1,650

Question 6 객관식

1189. Of the people surveyed in the age-group 65 or older, if the ratio of the number of women to the number of men was 3 to 2, how many women were in that age-group?
A.580
B.640
C.725
D.870
E.967

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 7 주관식

1190. Of the people surveyed who had long-term-care insurance, how many were age 55 years or older?

Question 8 객관식

1191. Which of the following is closest to the percent of all people surveyed who were in the age-group 45-54 and had long-term-care insurance?
A.4.4%
B.5.8%
C.7.5%
D.9.4%
E.12.3%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 객관식

855. The function h is defined by $h(x)=10^{(x+1)}$ for all integers x. Which of the following is equal to $(h(x+1)-h(x))/3$ for all integers x??
Ah(x)
B2h(x)
C3h(x)
D6h(x)
E9h(x)

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

757. On May 5, 2004, the closing price of Stock A was $60.00, which was a decrease of 8 percent from the closing price of the stock on the previous day, May 4, 2004. On May 6, 2004, the closing price of the stock was exactly the same as the closing price on May 4, 2004. What was the percent increase in the closing price of the stock from May 5, 2004 to May 6, 2004?
Give your answer to the nearest whole percent.

[ ] %

Question 11 복수선택

1162. Carolyn took out a one-year loan for $15,000 at 8 percent simple annual interest. She repaid the total amount, including the interest, by making 12 equal monthly payments on the last day of each month beginning in January. At the beginning of which of the following months did Carolyn have less than $10,000 of the total amount left to the repaid? Indicate ALL such months.
A.April
B.May
C.June
D.July
E.August
F.September

Indicate all that apply. [A] [B] [C] [D] [E] [F]

Question 12 객관식

608. In a certain apartment complex, the probability that a household selected at random will be one that subscribes to newspaper T is 0.5 and the probability that a household selected at random will be one that subscribes to newspaper W is 0.3. If the probability that the selected household will be one that subscribes to both newspaper T and newspaper W is 0.1, what is the probability that the selected household will be one that subscribes to neither newspaper T nor newspaper W?

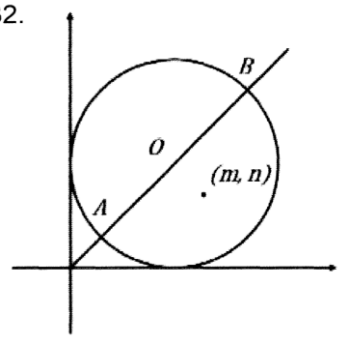
A0.7
B0.6
C0.5
D0.4
E0.3

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

1132.



In the xy-plane, a circle with point O at the center is tangent to both the x-axis and the y-axis. AB is a diameter of the circle.
Quantity A: m
Quantity B: n

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

519. When the total cost of 90 dollars for the rental of a van was evenly divided among n passengers, the cost to each passenger was less than 23 dollars.
Quantity A: n
Quantity B: 5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

1033. $x^2+y^2=25$. Both x and y are integers and x>y.
Quantity A: x
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

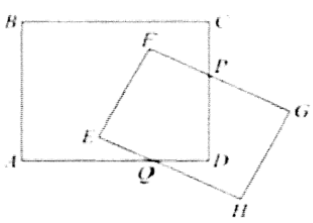
Question 4 대소비교

1301. Of the 70 students who took a typing test, 48 made 5 or more errors and 38 made 10 or fewer errors.
Quantity A: The number of students who made from 5 to 10 errors
Quantity B: 16

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

944.



In the figure shown, ABCD and EFGH are rectangles.
Quantity A: The sum of the measure of angle CPG and angle AQH
Quantity B: 300°

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

827. A flat rectangular garden has a diagonal that measures 10 feet. If the length of the garden is 3 times the width of the garden, which of the following is closest to the width of the garden, in feet?
A3
B4
C5
D6
E7

(A) (B) (C) (D) (E)

Question 7 객관식

975. If m is the median of fifteen consecutive even integer, then the greatest of these fifteen integers is
Am+12
Bm+14
Cm+16
D2(m+5)
E2(m+7)

(A) (B) (C) (D) (E)

Question 8 객관식

1256. When the positive integer x is divided by 42, the remainder is 19. What is the remainder when x is divided by 7?
A.0
B.2
C.3
D.4
E.5

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

796. The distance from Karen' s home to an office is 98.4 miles along a certain route. Along the route, a rest stop is located 83.8 miles from her home and a police car is always within 0.5 mile of the rest stop. Along the route, less than 0.5 mile before the office, Karen' s car broke down. Which of the following values could be the distance, in miles, along the route between Karen' s car and the police car?
Indicate all such values.
A13.0
B13.5
C14.0
D14.5
E15.0
F15.5
G16.0
H16.5

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G] [H]

Question 10 주관식

843. A customer ordered 391 identical handbags and must choose a shipping plan. The shipper will use any combination of containers of sizes that each hold up to 20 handbags, up to 12 handbags, and up to 5 handbags. The shipping costs for these container sizes are $3, $2, and $1, respectively. At most one of the containers will be shipped holding less than the maximum number of handbags for that container. For the total shipping cost, the cost of using only the containers that hold up to 5 handbags is how much greater than the least possible cost?

$ [ ]

Question 11 객관식

480. Of the positive integers that are less than 25, how many are equal to the sum of a positive multiple of 4 and a positive multiple of 5?
A2
B5
C10
D11
E22

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 12 객관식

736. What is the least integer that can be expressed as a product of four different integers, each of which has a value between -5 and 4, inclusive?
A–60
B–120
C–180
D–240
E–500

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 13 복수선택

750. If a and b are integers satisfying the equation $a^2+b^2=145$, which of the following could be the value of a+b? Indicate all such values.
A20
B17
C13
D10
E4
F0
G-1

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G]

Question 14 객관식

859. If an integer is chosen at random from the integers between 101 and 550, inclusive, what is the probability that the chosen integer will begin with the digit 1, 2 or 3, and end with the digit 4, 5, or 6?
A0.02
B0.05
C0.10
D0.15
E0.20

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 15 객관식

1134. In the xy-plane, line segment RS is a side of a square. The coordinates of R are (2, 10) and the coordinates of the midpoint of RS are (7, 12). Which of the following CANNOT be the coordinates of a vertex of the square?
A(6, 0)
B(8, 4)
C(8, 24)
D(12, 14)
E(16, 4)

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ